2026 AI Strategy

2026 AI Strategy

Our 2026 AI Strategy accelerates merchant activation, amplifies revenue through smarter marketing, and equips the business with fast, trustworthy insights. We focus on AI where the impact is immediate and measurable: shortening merchant activation time, improving production excellence and reducing innovation latency across teams. This document describes the key work streams from the AI team across 3 domains:

1. Merchant Assistance AI. How to support our merchants with AI assistance tools to become successful and scale on POD business.
2. Supply Optimization ML. How to support our internal production and supply demand to optimize their operations
3. AI Platform. How to uplift our AI platform capabilities to support the above points

Merchant Assistance AI

AI in POD Today: Status Quo, Impact, and Strategic Implications

Over the past three years, the print-on-demand (POD) marketplace has entered a structurally different phase. Product creation has accelerated dramatically while GMV has stagnated or declined, resulting in collapsing efficiency across all merchant segments. Funnel metrics confirm this shift: while the time to reach key milestones (Setup, AHA, Activation, Scaleup) has improved, the percentage of merchants reaching each stage has declined year over year . For example, only ~1.14% of merchants now reach AHA within 30 days, and just 0.15% reach Scaleup, despite faster average progression times. This indicates that speed is no longer the primary constraint—product-market fit and differentiation are.

AI tools introduced in 2025 successfully reduced execution friction but did not improve commercial outcomes. AI image generation adoption remains low and volatile: only ~10–14% of merchants use AI images in a given quarter, and just ~2–5% of products are created with AI-generated visuals. Critically, 0% of the top 3,000 products by orders (1M+ orders total) use AI-generated images, while 96.4% rely on uploaded, merchant-owned assets. This demonstrates that AI image generation is largely absent from the winning loop and does not drive GMV, activation, or scale.

By contrast, AI tools that improve performance rather than creativity show strong, consistent adoption. The Listing Optimizer achieves >85% opt-in and publish rates, with >60% of products using AI-generated titles and descriptions and ~65–70% of merchants actively using the tool. These tools integrate directly into existing workflows, improve SEO and positioning, and deliver immediate, measurable value without challenging merchant taste or strategy.
Impact by Merchant Segment

- Regular Merchants / New Entrants. AI lowers the barrier to creation and increases experimentation, but it does not improve success rates. AI merchants consistently take longer to reach Activation and Scaleup than non-AI merchants (e.g., median Activation in Q4'25: 7 days AI vs 3 days non-AI). AI currently amplifies exploration, not conviction, resulting in more published products but no improvement in demand generation.
- Power Sellers (PWS) face growing oversupply and declining efficiency. AI creation tools provide limited incremental value, while performance-oriented tools (SEO, catalog optimization) align better with their needs. Without stronger guidance on what to prioritize or stop, AI risks accelerating low-return catalog expansion.
- Super Power Sellers (SPWS) generate ~68% of top-performing products and dominate GMV, yet largely bypass creative AI tools. Their needs center on scaling complexity—personalization, embroidery, catalog management, pricing, and SEO at volume—areas where generic AI tools fall short but domain-specific AI can be highly differentiating.

Key Pains Identified

1. Fast creation, poor decision quality. Merchants can create and publish faster than ever, but most products fail to convert. AI reduces execution friction without improving what is created, worsening oversupply and GMV per product.
2. AI helps exploration, not commercial outcomes. AI-generated designs and assets are largely disconnected from commercial success. They are absent from top-performing products and do not improve AHA, Activation, or Scaleup rates.
3. High-value merchants are underserved by AI. The merchants who drive the majority of GMV see little value in current AI offerings, as these tools do not address scaling, optimization, or complex product workflows.
4. The catalog is generic, opaque, and not growth-oriented. Today, the FYUL catalog is not personalized, not inspirational, and not growth-focused. All merchants see largely the same products and rankings, regardless of niche, performance, or intent. Recommendations are mostly static and human-driven (e.g., bestsellers), lacking algorithmic ranking, trend responsiveness, or behavioral adaptation.

2026 Merchant AI Initiatives

The current data makes one conclusion clear: AI should not be positioned as a creativity or "business-building" engine for new POD entrants. That promise is no longer sustainable in a saturated market. **The biggest unlocked opportunity lies in repositioning AI from a creation tool to a decision intelligence layer that guides merchant growth**:

- Personalizing discovery based on merchant behavior, niche, and performance
- Surfacing evidence-based growth and diversification opportunities
- Guiding merchants through complex choices (products, variants, providers, decoration methods)
- Supporting high-leverage workflows such as personalization, embroidery, SEO, and portfolio optimization

Shifting AI investment toward these domains aligns with actual merchant adoption, supports high-GMV segments, and directly addresses the structural inefficiencies now limiting marketplace growth.

We are shifting from "making things easy to create" to **"Here is a business opportunity, here is the evidence why, and here is how to launch it in one click."**

Pillar 1: The Commercial Decision Engine (The Brain)

This is the logic layer that powers the platform's "Business Intelligence." It processes market signals and merchant performance to create Explainable Recommendations. It doesn't just suggest a product; it provides a Product Evidence Card showing margin potential, market fit (e.g., "Top 20% on Etsy EU"), and risk signals (return rates).

Why it matters
The 1.14% AHA rate proves merchants fail because they lack data. By making recommendations "explainable," we build trust. When a merchant sees *why* a product is recommended (e.g., "Fills a gap in your catalog"), they move from "exploring" to "executing" with conviction.

Success Metrics

1. **[Primary] AHA Improvement Index:** % increase in merchants reaching AHA within 30 days. (Purpose: Measures if evidence-based guidance leads to faster product-market fit).
2. **Evidence Trust Score:** % of merchants who click "Why this product" and subsequently publish that item. (Purpose: Validates that transparency drives action).
3. **Predictive Margin Accuracy:** The delta between AI-predicted margin potential and actual merchant profit. (Purpose: Ensures "Business Evidence" is financially sound).

Roadmap

- **H1 2026:**
    - **(Q1) Merchant Growth Profile Service.** Build the internal model that categorizes merchants by niche, channel, and maturity.
    - **Design Recommendation.** Launch the "winning style" extraction engine. Connect it to the creation flow so merchants start from a trend-aligned prompt rather than a blank page.
    - **Explainable Recommendations.** Launch the "Recommended because..." logic. Connect the Growth Profile to the catalog to explain *why* specific items appear.
- **H2 2026:**
    - **Product Evidence Cards.** Full rollout of business snapshots on catalog tiles (Margin, Market Fit, Risk signals).

Pillar 2: Professional Scale & Complexity

This pillar automates the high-value workflows that Power Sellers (SPWS) struggle to scale. It includes **Build-a-Collection**, which uses AI to apply a single design across a cohesive set (e.g., "Festival Pack"), and **Technical AI** for embroidery and personalization. It also includes **Design Templates** that are pre-optimized for conversion.

Why it matters
High-value merchants drive 68% of GMV but avoid "creative" AI. By solving "Technical AI" problems like embroidery digitizing and personalization layers, we create a moat that generic AI platforms cannot match. This reduces the rejection rate for complex items and increases the variety of high-margin products in the marketplace.

Success Metrics

1. **[Primary] Catalog Scaling Efficiency:** GMV ÷ number of active listings. (Purpose: Measures if merchants are scaling with *quality* rather than just flooding the catalog).
2. **Collection Adoption Rate:** % of merchants publishing "Bundles" or "Sets" via the AI guided-flow. (Purpose: Measures shift from single-item to brand-building).
3. **Technical Success Rate:** Reduction in digitizer rejections for embroidery and personalization errors. (Purpose: Measures the "print-readiness" of AI-assisted outputs).

Roadmap

- **H1 2026:**
    - **(Q1) Auto-Background Removal & Image2Image Personalization.** Creating designs involve multiple manual steps. Auto-background removal will workflow by automating the repetitive tasks of cleaning assets and setting up customizable text/image layers. Fix the core friction in the creation flow for customizable products.
    - **Embroidery-Friendly AI** + **"Build a Collection" .** Launch the guided flow to turn a single design into a "Merch Pack" (Tee + Hoodie + Tote) in one click.
- **H2 2026:**
    - **Design Templates (4.2)** + **Portfolio Guidance Models.** Launch one-click design templates categorized by niche (e.g., "Fitness Memes") that are proven to convert on specific channels.

Pillar 3: Growth-Aware Algorithmic Marketplace

This pillar moves the FYUL catalog from a "static list of bestsellers" to a "personalized opportunity feed.". This is the "front door" of the marketplace. It replaces static banners with Mockup-Driven Inspiration —showing products in lifestyle scenes (e.g., "Streetwear looks your customers love") rather than blank SKUs. It also includes the New Arrivals Engine to ensure new providers and products get data-driven exposure.

Why it matters
The catalog currently feels like a warehouse, it’s "generic and opaque."; it should feel like a trend-gallery. By leading with **Mockups** and **Social Proof** (e.g., "Used by 37% of similar stores"), we reduce the cognitive load of browsing and inspire merchants to expand into new categories they hadn't considered.

Success Metrics

1. **[Primary] Discovery-to-Publish Conversion:** % of catalog views that result in a published product. (Purpose: Measures if personalized ranking is surfacing relevant items).
2. **Niche Expansion Rate:** % of merchants selling in a 2nd or 3rd category after seeing an AI "Inspiration" view. (Purpose: Measures the impact of the "Idea Generator").
3. **New Product CTR Lift:** The increase in engagement for new blanks/printing methods via the New Arrivals Engine. (Purpose: Measures the platform's ability to diversify the supply chain).

Roadmap

- **H1 2026:**
    - **(Q1) New Arrivals Engine** Move away from ad-hoc exposure. Use a controlled exposure period for new blanks/providers to gather data and automatically promote "winners" to relevant merchants.
    - **Semantic Search.** Implement "intent-based" search
    - **Customized Ranking v1** + **Mockup-Driven Inspiration.** Replace bestsellers with "For You" rankings and launch lifestyle-first catalog views.
- **H2 2026:**
    - **Social Proof Benchmarks** + **AI Banner/Promotion Management.** Automatically manage catalog real-estate to show merchants the promotions most likely to drive their specific growth.

Supply Optimization ML

Supply ML pain points

Here, we have a list of pain points that our team plans to look into in 2026, through the planned initiatives:

Forecasting logic

Lack of external inputs in forecasts

At the moment, the forecasts are generated based almost solely on historical data, which will not always give us a perfect picture of what to expect in the future. Especially when moving into

Enterprise segment, prior notices will be extremely important when it comes to stock and employee planning

Problem we need to solve
We need to add a way to get and use external inputs into forecasts, so they are no longer based solely on historical data, so we can better anticipate spikes in demand before they happen

No forecasts for supplies

For our 1P business, we need to buy not only stock products, but supplies to make it possible to produce these items as well. Practical examples include inks for DTG printers (or stickers), colored threads for our embroidery machines, or the raw material for our cut&sew products. At the moment, there is no reliable forecast for these supplies, which puts additional strain on our purchasing team and does not allow us to do timely automated buying of these products

Problem we need to solve
Need to create forecasts for key supplies so that their purchasing can also be automated

External dependency on 3P forecast generation

Currently, a key input needed for our 3P (and, by extension, the volume transfer part of 1P) forecasts is the Printify Choice adoption rate for the upcoming period, which is created by the Strategic Finance team. This is something we need to take over on our side to enable faster forecast generation in cases where the forecasts either need to be generated again for some reason, or for when we switch to a weekly forecast generation schedule - we should not require manual effort from another team to generate an updated forecast

Problem we need to solve
Taking over calculation of as many of forecast inputs as possible, to make it possible to switch to a weekly forecast cadence in the future

Forecast generation is manual

We are limited to generating a forecast once per month due to the manual effort required to accomplish this. It is necessary to move toward automating the creation of forecasts to enable weekly updates, which would be important for our 1P purchasing

Problem we need to solve
To improve 1P purchasing accuracy, we need to switch to a weekly forecast cadence

1P operations improvements

Accounting for 1P volume smoothing

In 2026, our 1P team wants to focus on smoothing the incoming 1P order volume throughout the year, meaning that we want to have a smaller difference between volume received in

non-peak and peak seasons of the year, to simplify employee and stock planning. There is currently no agreed process on how this needs to be implemented, which will be a key talking point in early Q1

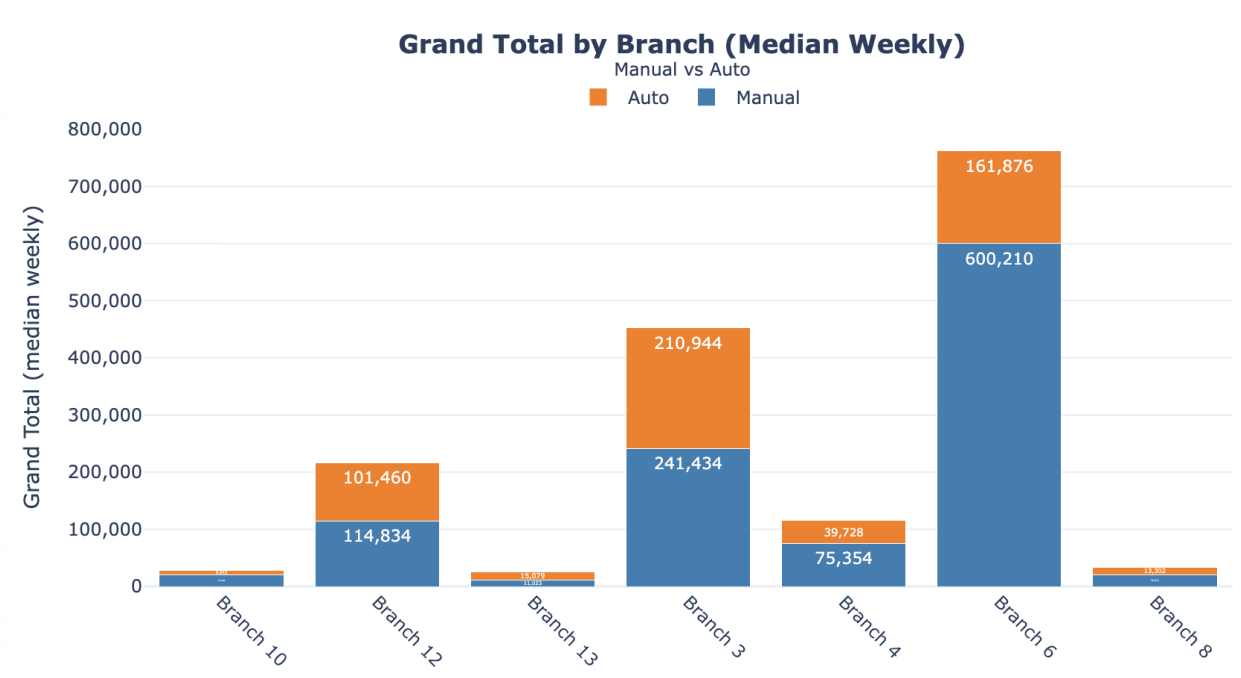
Problem we need to solve

Create a process for how we will deal with volume smoothing requirements from our 1P business, and ensure they are reflected in our forecasts for both 1P and 3P businesses

1P purchasing is mostly a manual process

In 2026, the expectation is that ML forecasts will be used as a basis for almost all purchasing decisions on 1P side. One part of those are automated weekly buys where ML data is used in limited capacity since early December of 2025. Using ML forecasts, a list of required products is created for each facility and goes through a process of automated or manual buys (based on whether all automated buying criteria are met)

Currently, the majority of the buying is still done manually due to specific items not meeting a threshold for automated buys and resulting in our purchasing team to get involved to make a decision on whether to combine some items into a single order, choose another supplier, or make a larger order to fulfill supplier requirements. Below, we can see an image showing the median weekly purchase volumes (in USD), showing the manual order and automated order proportion. The goal with our stock ordering project is to move more of the volume to be bought automatically



As ML forecasts are used as a basis for the calculations to determine what needs to be bought, we will also assist in creating a logic for shifting to a larger percentage of orders being made without manual input, freeing up resources on the purchasing team side

Problem we need to solve
Optimize the stock buying process by increasing the proportion of automatically generated orders, making the work easier for both purchasing team, as well as inventory team who can avoid buying schedules that are currently required to carry out these purchases

DTG color mode decisions are manual

Before printing any DTG item, it is necessary to provide information on the expected printing color mode (White only, Color only, Color + White) to the printer to ensure that the final print is as good as it can be. At the moment, determining the color mode is mostly outsourced to digitizers, which costs our company around 100k USD per year, makes production slower, as well as decreases our DTG quality in some specific cases

Problem we need to solve
We want to avoid having to use an external solution to determine color mode, as this is a rule based logic that we can automate to be equal to or better (in specific cases) than the manual inputs that we are spending 100k a year on currently

10x product onboarding

The process for product onboarding is complicated and very time consuming for our partners, requiring a lot of manual effort. To make this process smoother, we must look at automation opportunities for the steps our partners need to take to make this onboarding process more efficient. Some of the key issues at the moment (as listed HERE):

- Product Onboarding capacity is limited by team bandwidth rather than market opportunity, restricting catalog expansion.
- The lack of a true self-serve flow forces FYUL to act as intermediaries for data collection and validation.
- Operational workload grows linearly with product volume, making it difficult to scale without increasing headcount.

Problem we need to solve
We want to assist with the automation of the product onboarding flow, to ensure that our partners can onboard their products faster and provide a larger choice to the end users

2026 Supply ML initiatives

Initiative 1: Improvements in forecasting logic

In 2026, we plan to continue work on improving and creating critical functionality related to our forecasting efforts. This will involve more work with our 1P and 3P stakeholders to ensure the expected volume splits between our businesses (more on this - under initiative 2, below), as well as a focus on dealing with various outliers and automating the generation of the forecasts,

which are currently created manually. For this, we have already identified multiple projects that will be key for our team throughout 2026:

Manual inputs and other outliers in forecasts - H1 2026

This project will deal with adding manual inputs from 1P business side to our forecasts, as well as analysis on identifying specific outliers (as an example - the proportion of bulk orders in a specific month), which will help us in two ways:

1. If manual inputs are provided in a timely manner, it will allow us to plan for the necessary blank purchases and allocation of staff to specific areas of our 1P business to cover the expected surges in volume
2. Identifying and labelling this outlier data in historic volumes will allow us to create a better “baseline” forecast, which is a version of forecast without any external inputs, to which we can then add any required modifications. As the number of various external inputs keeps increasing (which is to be expected with ML forecasts being used for decision making purposes in more parts of our business), it is critical to ensure that the baseline forecast version is as close to reality as we can make it. This is because there are some steps we then make to split the expected volume proportionally both on 1P and 3P side of the business, so a wrong baseline will mean that we introduce a base error in our forecasts even before applying various manual inputs (which will have their own error rates)

Taking over Printify Choice adoption rate calculation - H1 2026

Another key step we want to make is to try and ensure that we can calculate as many of the needed inputs for ML forecasts as possible, a key point here being PC adoption rate calculations. These are used to split volume correctly between our 3P vendors, and to be able to estimate Volume Transfer to our 1P business

Currently, the adoption rate is calculated by the Strategic Finance team, but we plan to take over these calculations (for everything except unreleased products) on our side - this will be important for when we switch to a weekly forecast cadence, as well as in cases where we need to generate an additional version of the forecast for any reason and need the most up to date information

Expanding the ML forecast coverage on 1P and 3P side - H1 2026

This will be handled in two parts:

1. 3P side forecast coverage will be increased by moving to use the FYUL category tree as a basis for our categories, as opposed to the now outdated ex-Printify category tree. This is planned for early in Q1 and will increase our 3P forecast coverage from around 90% to closer to 97% of products
2. 1P side forecast coverage will be increased by creating forecasts for some of the supplies that we need to buy to support our product creation. An example here are inks for our DTG printers, which do not have proper forecasts at this point in time. The plan here is to work our way through the supply one by one (in order of importance to our

business) and expand the forecasting logic to also include these items, and help in making automated purchasing decisions for supplies (which we will also be assisting with through initiative 2, below)

Automating the generation of forecasts - H2 2026

The ultimate goal of our ML forecasting model is to ensure that it can be run automatically, with as little manual input as possible, and this is what we are working towards with the various features that are being implemented. Automating forecast generation will allow us to switch to a weekly forecast cadence (as opposed to monthly), and this would help us to quickly react to any unexpected changes in demand faster than we do now. It will be especially useful on 1P purchasing side, to ensure that we don't over or understock with specific items

Generating automated forecasts is an ongoing project since Q4 of 2025, and is expected to be completed in H2 of 2026 as it requires all functionality from this initiative to be completed, among other technical improvements

Initiative 2: Operations process optimization

In addition to focusing on expanding the forecasting related functionality, our 2026 plans also include working closely with our 1P operations teams to assist them in optimizing various processes, some of which are outlined in more detail below. This is not a complete list of projects we expect to tackle, but are instead a look into some of the work we have already begun discovery on by the end of 2025

Implementing 1P volume smoothing in forecasting - H1 2026

As part of this project, we will work with 1P stakeholders to understand in-house team short and mid-term plans and volume expectations, and then ensure that these are reflected in our 1P and 3P forecasts, as well as calculate the necessary changes on order routing team side to ensure that this result then is met in practice and our 3P vendors are aware of what will happen

The expected impact here is that we will want to get additional 1P volume during the non-peak period of the year, and give this additional volume away during the peak season. While this is a relatively straightforward task on the 1P side, SupplyML team will work with our 3P stakeholders to calculate the expected impact on our 3P vendors as this additional 1P volume will have to come from somewhere. Since this will not be a “natural” order split anymore, we will also calculate what changes would need to be made on the order routing side (on a blueprint + print provider level, most likely) to ensure that we get the expected results

This is also where a good baseline forecast will be important, as we can likely expect the volume smoothing requirements to change over time, and would still need to know the “natural” routing outcomes to be able to calculate what changes would need to be made on order routing side for any future changes in volume smoothing needs

Optimizing the stock buying process - H1 2026

SupplyML will create a logic that can automate the first two of the options purchasers do - making it possible to automatically combine multiple items into an order to the best supplier for a combination of the products (as opposed to looking at each line item separately), as well as consider alternative suppliers for specific cases. The expected outcome here is a larger proportion of automated buys of our products, requiring less manual input from our purchasing team side

This will also be helped by one of our initiative 1 projects - generating forecasts for supplies. At the moment, there is no trusted source for supply product future needs and our purchasing team needs to use their experience and judgement to determine how much needs to be ordered for each facility. Once we can generate initial forecasts, it will be possible to automate these purchases as well

Determining the correct DTG color mode - H1 2026

Our plan here is to work with R&D team and create an automated solution for determining the correct color mode for our DTG products. There has been a past initiative that has tackled parts of this, which will be analysed to see whether we can use parts of it in the final solution

This is not expected to be a very resource intensive project on SupplyML side, but is one with a very clear set of benefits:

1. DTG print quality for specific cases will be improved and remain at least the same in all other cases (we know that digitizer outcomes are less than ideal for multi-placement items)
2. DTG production time will be decreased, as we will not need to wait for digitizer response before starting the production process
3. Decreased reliance on outside services for our production
4. Expected savings of 100k USD per year

Initiative 3: Supporting the 10x project

The 10x product onboarding project is one of the main projects for FYUL as a company, and the SupplyML team will do its best to support these efforts wherever possible. In addition to that, we will also continue our work on expanding the use of our churn/downgrade model to ensure that merchants keep using our platform

10x product onboarding - H1 2026

For this project, we have currently identified two use cases that could require ML team involvement, both of which will be investigated further to understand their technical feasibility before spending significant time on development

The first of these projects is about layout generation - when a merchant adds a new product, we need to create its virtual layout (including print areas) so that it can be used when users want to create a new design on this product. The process to create a layout is currently fully manual and

takes a lot of time. If possible to automate, we could save significant time on new product creation for our vendors. Discovery work on this task is scheduled for the beginning of Q1, to better understand the scope of the project, as well as the technical requirements from our stakeholders

Another somewhat related project deals with a step before this takes place, and that is product import. Here, we would need to create an import tool that can take a product information file with different data layouts and column names, and automatically map it to the internal product intake sheets - listing missing data that could then be added by the respective vendor. Currently, there is manual mapping for some of our largest vendors, but the process is fully manual for all other vendors. This is a project we expect to revisit either in late H1 or early H2, to do some additional discovery work and understand the feasibility of an automated import tool

We will also work together with the relevant teams to look for other automation options within the product onboarding process

Initiative 4: Discovery work for future projects

Throughout the year, we plan to work with our stakeholders in determining other possible development areas for the future. Some examples are listed below, but this is in no way a complete list

1P employee shift planning

Since we are providing 1P planning forecasts which are used for this purpose, it makes sense to do the next step here and provide suggestions on the necessary number of employees in each facility for each technique based on the ML forecast

This could be further expanded by also suggesting the actual employees to fill these spots on a daily/weekly basis based on employee skills, availability, as well as other constraints

Pickup shelving usage optimization

During peak season, we have had issues with not having enough available space in the pickup locations, resulting in an inability to fulfill some orders in time as there is no way to move specific items from excess stock to a pickable location

We could create a logic that continuously monitors incoming orders and optimizes pickup shelving usage to be able to fulfill all critical orders. This would also mean optimizing the item locations themselves, ensuring that items that tend to be picked in a single pickup filter are located near to each other to speed up order pickup

Additional stock buys based on space availability

If we know how much space is available in each facility, based on the ML forecasts we could make larger purchases of specific items if that can lead to lower pricing, as we would be able to see that there is enough space for these items in that specific facility

Similarly, having this data available could help with better peak season preparation as it would be possible to anticipate expected space usage per facility and potentially lead to a decision to stock some products in just one facility in the region to save space in other facilities

AI Platform

FYUL's ambition to support merchants with intelligent, end-to-end AI assistance is constrained today by existing platform gaps. The 2026 AI Platform roadmap is organized around addressing these core pains through three tightly integrated platform initiatives.

Core AI & ML Platform Pains

Pain 1: AI Operates Without Real-Time Merchant Context

Today, there is no unified system that captures and exposes low-signal, high-frequency merchant behavior (design edits, blueprint usage, browsing patterns, product configuration decisions). Models rely largely on batch data and partial signals, causing them to make generic or mistimed recommendations. This prevents personalization, weakens reasoning, and limits the models ability to guide merchants effectively in the moment.

Pain 2: AI Lacks Native Understanding of POD Domain Logic

Fyul's current AI capabilities do not deeply encode POD-specific constraints such as print areas, aspect ratios, AOP behavior, or catalog structure. Knowledge is fragmented across services and not consistently available to LLMs. As a result, recommendations are often inaccurate, require manual correction, and fail to scale across product creation, optimization, and portfolio management workflows.

Pain 3: Legacy ML infrastructure

30% of AI services such as Prompt Enhancement still rely on legacy SageMaker endpoints, leading to higher operational overhead, inconsistent observability, slower incident resolution, and duplicated tooling across teams. This fragmentation increases cost, complicates ownership,

and prevents the AI Platform from enforcing consistent standards for deployment, monitoring, and model governance—ultimately limiting scalability and trust in production AI workloads.

Pain 4: ML Platform Friction Slows Innovation and Increases Cost

The team still operates in an operational silo for deploying new services and modifying existing webservices, largely because these processes are not automated end-to-end. Service changes often require manual code changes which are ideally not needed, increasing both cost and operational risk. As a result, iteration speed is reduced, confidence in AI outputs suffers, and the platform's ability to scale is constrained.

2026 AI Platform Initiatives

Initiative 1: Low-Signal Aggregation & Real-Time Context Layer

To address the lack of real-time context, Fyul will build a Low-Signal Aggregation Layer that continuously ingests merchant and product events, constructs state machines, and auto-generates features for LLMs and predictive models. This layer provides live context windows that allow AI to understand what a merchant is doing, what they prefer, and what actions are most relevant next.

Initial Use Cases will be about Growth- and portfolio-based personalization, which will be enabled by this Initiative.

Success metrics:
Event retrieval latency <1s, feature coverage per merchant/product >95%

Initiative 2: FYUL LLM Foundation Layer

To address domain understanding gaps, FYUL will leverage a shared AI foundation that combines large language models with POD-specific knowledge sources such as constraints, catalog semantics, and merchant state.

On top of this shared foundation, task-oriented capabilities (e.g. product creation, portfolio optimization, and scale guidance) can be implemented in a modular way, enabling consistent reasoning across multi-step workflows. By making the domain knowledge consistent and unified, models are aligned with intelligence and user behavior, reducing fragmented implementations, manual corrections, and inconsistent outputs.

Success metrics:

> 90% accuracy on domain-specific reasoning tasks, strong confidence calibration, measurable reduction in manual corrections.

Initiative 4: MLOps Platform Reliability, Performance & Simplification

To remove platform friction, the roadmap includes automating the discovery and publishing process for new web services. We counted an average of 5 days of delay per service that should have been published in Q4 and 3 days per month of work spent on updating and creating new public endpoints on the API Gateway. In Q1 we planned to have 14 such modifications, which will likely not yet benefit from the new API Gateway Auto Discovery and thus cost us roughly 14 days of work and the delays associated with it.

Auto Discovery for API Gateway

Automatically retrieve, validate, and merge OpenAPI specifications into the API Gateway, enabling data scientists to publish new model endpoints without manual MLOps intervention. This creates a self-serve, low-friction deployment flow that accelerates experimentation, shortens iteration cycles, and reduces operational bottlenecks.

Remove legacy Sagemaker endpoints and migrate to AI/ML Platform

In our year-end survey 2025, the team identified the legacy SageMaker endpoints as a source of knowledge silos. By migrating the Prompt Enhancement service to the unified AI/ML Platform on NCP, these silos will be eliminated and ownership will become more transparent.

In addition, during our 2024 survey we also interviewed members of the SRE team, who highlighted the operational overhead associated with managing SageMaker endpoints. This migration addresses those concerns by reducing operational complexity, improving observability and reliability, lowering costs, and ensuring consistent tooling, monitoring, and model governance across all AI workloads.

Success metrics:
AI Platform inference requests served, 5xx error rate, service uptime, % of models migrated to NCP, total AI platform cost.

AI Platform Metrics

According to our roadmap our focus is on impacting the following metrics:

KPI Title	Definition
AI Inference Requests Served	Monthly Total of Inference requests

Our health metrics look as follows:

KPI Title	Definition
Service 5xx Error Rate	Measuring the 5xx error rate is crucial because it indicates unexpected server side issues that directly affect the end users experience.
Service uptime	Service uptime is not yet measured, our services are expected to be up all the time, but we want to have additional assurance that tells us that they are actually reachable.
Services/Models not on NCP	How many models did we not yet migrate to NCP, this shows us how many models still need to be moved to NCP and use our standardized toolkit.
AI Cost	The AI Platform cost is substantial in comparison to other cost items, this is the piece that should be reported on across various platforms.

Appendix - Merchant data evolution

Appendix.

2025 Merchant Insights

Across all merchant segments, the last 3 years show one universal pattern: Product creation is growing exponentially while GMV is flat or declining. Marketplace efficiency is collapsing. This is not a seasonal fluctuation, it's a structural imbalance between supply growth (massively accelerating) and demand growth (flat or shrinking in some segments). Our qualitative research shows that POD merchants are constrained by limited time, fragmented workflows, and gaps in know-how required to succeed in a competitive print-on-demand market.

In 2025, FYUL delivered several AI capabilities that addressed isolated pain points — from image generation and trend-based product creation to listing optimization and creative ideation. These tools proved valuable but operated as disconnected solutions. In 2026, we will expand on the foundation established with the Co-Founder Agent and evolve toward a fully integrated, end-to-end agentic AI system that guides merchants from identifying product opportunities all the way through creation, scaling, marketing, and ongoing performance management. This shift transforms our AI efforts from feature-level enhancements into a unified, intelligent growth engine for every merchant segment.

Product creation is growing faster than GMV

Across the last three years, merchant behavior shows a consistent pattern: product creation is growing faster than GMV, leading to declining efficiency across all segments. The entire marketplace is producing more products with declining returns, signaling that merchants lack guidance on what to create, how to optimize, and how to scale efficiently — pointing directly to the need for AI-driven creation, prioritization, and growth support.

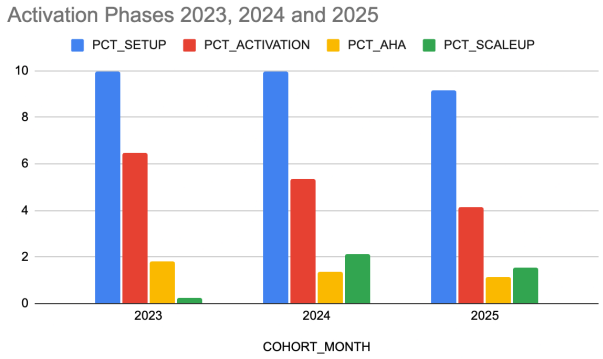
- **Regular Merchants — High Volume, Low Efficiency, Marketplace Oversaturation.** Regular merchants produce the largest share of catalog growth, creating ~8.5M products annually but generating only ~$116M GMV ***($13.64/product)*** While the merchant base continues expanding rapidly, their efficiency collapses: GMV per merchant and per product have steadily declined, and the majority of newly created listings fail to convert. This segment drives massive oversaturation, with millions of low-quality or low-fit listings contributing minimal revenue. The average number of products created per merchant has decreased -28% from 2024 to 24th November 2025.
- **Power Sellers (PWS) — Growing Supply, Stagnating Demand.** PWS create ~2.5M products/year and generate ~$72M GMV (***$28.8/product***). Although their supply has grown, GMV has stalled, and efficiency metrics (GMV per product, GMV per merchant) are trending downward. The segment remains healthy but is now clearly facing overproduction relative to demand. Without intervention, it risks moving toward

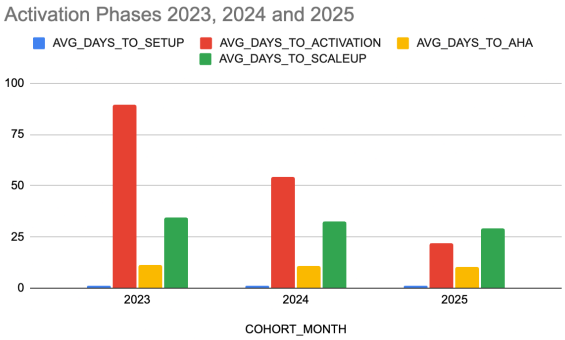
Regular-segment inefficiency. The average number of products created per merchant has decreased -21% from 2024 to 24th November 2025.

- **Super Power Sellers (SPWS) — High-Value Engine Showing Early Decline.** SPWS produce ~5.8M products/year and generate ~$186M GMV (***$32/product***), remaining FYUL's strongest driver of revenue. Yet even here, efficiency has fallen sharply: GMV dropped 25% in 2025 despite continued growth in product creation. GMV per merchant and per product are declining year over year. This segment is still highly valuable but showing early symptoms of saturation and productivity decay. The average number of products created per merchant has increased 9% from 2024 to 24th November 2025.

Merchant activation and product success are declining Year-over-Year

Cohort analysis across Setup, Aha, Activation, and Scale-Up shows a clear and persistent deterioration in new-merchant success. Although merchants who *do* reach key milestones are doing so faster than ever, the *number* and *percentage* of merchants who reach those milestones continue to fall. This reveals that the early journey still contains substantial friction in ideation, design quality, listing performance, and go-to-market readiness.





Across every milestone, we see the same pattern: efficiency is improving, but activation volume is collapsing. Merchants are technically able to progress faster, yet fewer of them ever make meaningful progress — a sign of deeper gaps in guidance, product-market fit, and early-stage support.

- SETUP (connected store + 3 published products within 7 days) has decreased slightly from 1.32 → 1.16 days (faster). Merchants are reaching SETUP extremely quickly once they engage — essentially same-day or next-day — but fewer merchants are reaching this stage, dropping from ~531k → ~405k in two years. This suggests that the primary friction is not speed, but getting merchants to start and complete the early creation/setup workflow, especially around idea generation, design confidence, and understanding what to publish.
- AHA (first sale via integration within 30 days) has improved from 11.49 → 10.31 days. The number of merchants who achieve an early sale is shrinking even faster than SETUP. Only 1.14% of merchants now reach AHA within 30 days — down from nearly

2% two years earlier — despite the process becoming slightly faster for those who succeed. This indicates weak early product-market fit, ineffective listings, poor differentiation, or lack of trend awareness, causing most merchants' early products to fail to convert.

- ACTIVATION (first successful order sent to production) has improved dramatically from 89.6 → 22.0 days. However the % activated merchants is becoming much rarer (6.46% → 4.13%), even though those who do activate do so dramatically faster. This widening gap highlights that speed is not the problem — product success is. Most merchants fail to reach any commercial traction, signaling early issues with product quality, relevance, pricing, and audience discovery.
- SCALEUP (10 orders via integration within 60 days) has improved from 34.3 → 29.1 days. Scale-Up remains extremely rare and is declining further: from 0.23% → 0.15% of merchants. Even among merchants who activate, only a small and shrinking group is able to scale products to repeat demand. This suggests structural challenges such as market saturation, weak catalog iteration, and low visibility for new sellers.

The insights of these 3 years can be summarized as:

1. **Funnel leakage at every milestone**. Percentages decline year over year for SETUP, AHA, ACTIVATION, and SCALEUP. Fewer merchants are advancing at every stage of the journey.
2. **Efficiency is improving but outcomes deteriorate**. Across all milestones, average days decrease, meaning successful merchants move faster. But the share of merchants reaching each milestone is collapsing, signaling that speed improvements do not address the real friction.
3. **Early-stage friction remains the primary blocker.** Merchants fail early due to issues in: idea generation, design quality and relevance, listing optimization, audience and trend understanding, creating products that can actually sell.
4. **Product-market fit is the dominant challenge.** The sharp decline in AHA and ACTIVATION rates shows that even when merchants publish products, they struggle to generate demand.
5. **Scaling is increasingly rare.** SCALEUP is now achieved by just **0.15%** of merchants, indicating an environment where very few products gain enough traction to grow.

Merchant and FYUL Are Not Benefiting from Product Catalogue and Diversification

Across the last three years, merchants and the platform are underleveraging the product catalog, limiting growth and diversification opportunities. Merchants struggle to discover high-potential products due to generic search, limited filters, and overwhelming choices. The catalog does not provide actionable guidance or relevant inspiration, leaving merchants uncertain about what to create next, how to optimize their offerings, and how to grow revenue. Recommendations are often perceived as irrelevant, repetitive, or opaque, with key metrics such as "sales potential" or portfolio gaps unexplained, reducing trust and adoption. From FYUL's perspective, new products and printing methods suffer from inconsistent exposure,

delaying validation of high-potential items and reducing overall efficiency. Operational friction, including heavy reliance on engineering for promotions and the limitations of a rigid, outdated taxonomy, further constrains catalog personalization, cross-selling, and marketing agility. This combination of merchant uncertainty and platform inefficiency has resulted in declining returns on new product creation: GMV generated by new articles over the past three years fell from $22.7M in 2023, to $29.8M in 2024, and dropped sharply to $7.4M in 2025.

Merchants face predictable and recurring frictions

Qualitative insights support the above quantitative data. Across segments, merchants face predictable and recurring frictions that fall into three major categories: profitability uncertainty and workflow inefficiency. These frictions reduce activation, growth, and GMV efficiency.

- Profitability & Pricing Uncertainty. Merchants — especially new or international ones — struggle to understand costs, set profitable prices, and anticipate margins. Shipping costs, product pricing, and cross-platform fees are hard to consolidate, creating financial ambiguity that blocks both activation and scaling. For larger merchants, inconsistent profitability and multi-item shipping penalties reduce trust and growth potential.
- Workflow & Operational Bottlenecks. From creation to fulfillment, merchants encounter time-consuming, error-prone workflows: manual product creation and listing slow down catalog expansion; personalization workflows break at scale and often require third-party tools; syncing issues, price updates, and variant management consume hours per week; personalization, mockup creation, and order management require repetitive manual steps. These bottlenecks disproportionately affect PWS and SPWS sellers who operate at scale.
- Knowledge & Direction Gaps. Inexperienced merchants lack the know-how to: identify winning product concepts; price, market, or position their products & choose trends, niches, or listing strategies. This results in high abandonment, underpricing, and low product performance.

There are other structural barriers, specially for International merchants (India, Pakistan), that are beyond the scope of this AI strategy document. Like compliance obstacles preventing store connection; high startup costs; limited access to key platforms (e.g., PayPal, Shopify activation issues)...

Merchants are not fully leveraging AI tools

In this recent Q3 2025 FYUL study among 812 merchants, 62% merchants have already incorporated AI into their daily workflows and they value AI as a tool for efficiency rather than a creative replacement. The main areas of use are: 57.6% merchants are using AI for SEO and product descriptions; 27.3% of merchants for design ideation and 11% for marketing and ad support.

Merchant product creation and GMV trends (24th Nov)

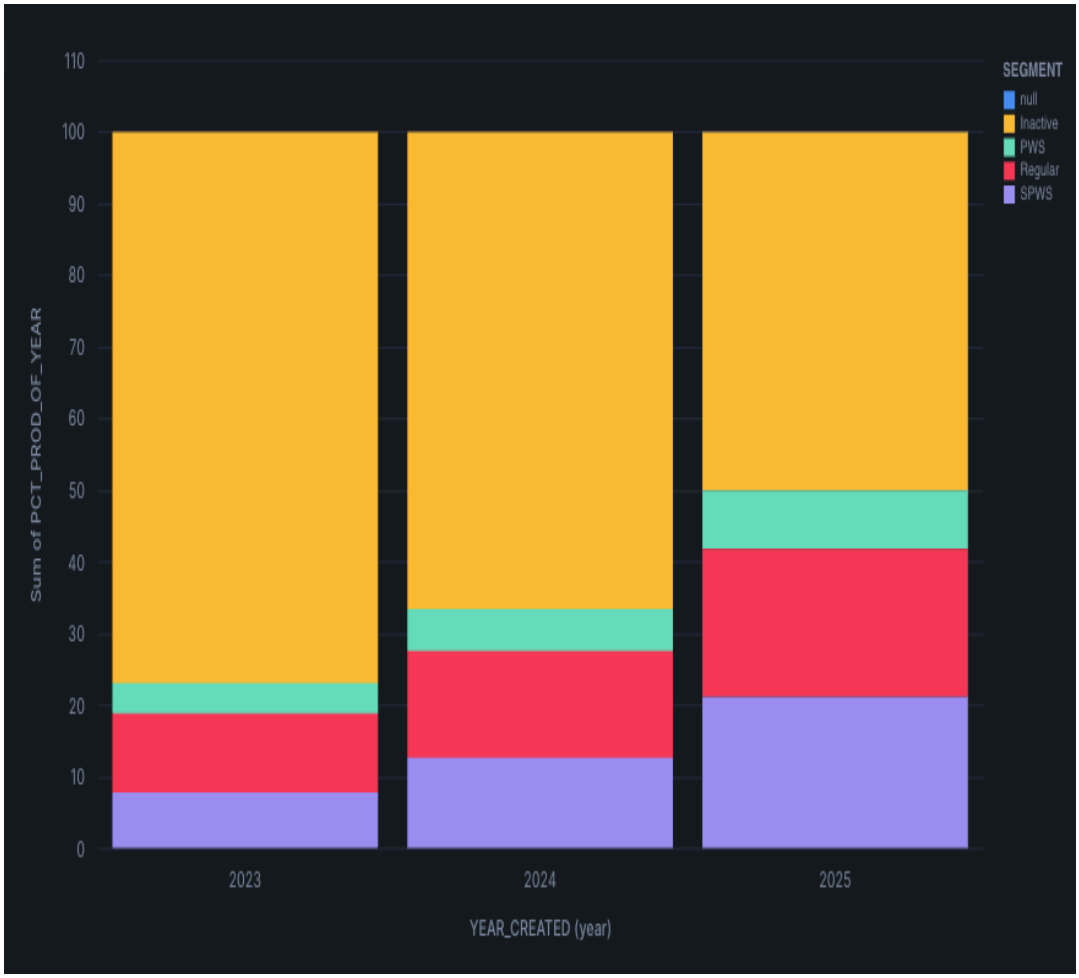
Analyzing the evolution of GMV and product creation during the last 3 years show this data
Data for 2025 was generated in 24th Nov. Analysis may need to be redone after peak season to confirm some of the trends.

Merchants can be broken into the following segments based on the GMV. The observed patterns of the production creation and the GMV during the last 3 years can be summarized as:

- Regular Segment. Regular merchants are experiencing severe efficiency decay. They created ~8.52M products/year (~710k/month) and generated ~$116M GMV/year (~$9.7M/month). However, merchant growth is disproportionately high—especially in 2025—while GMV declines. GMV per merchant fell from $272 → $297 → $284 and GMV per product dropped from $19 → $15 → $10. This segment is the main source of marketplace oversaturation: huge inflow of low-performance sellers creating millions of products that barely convert into revenue.
- PWS (Power Sellers) shows growing supply but stagnating demand. Over the years, they created ~2.48M products/year (~206k/month) and generated ~$72M GMV/year (~$6M/month). However, GMV peaked in 2024 and fell in 2025 despite more merchants and more products, leading to declining GMV per product (from ~$28.7 → ~$22.1) and flat GMV per merchant (~$2,900–3,000). This indicates oversupply: merchant volume and product creation are outpacing demand. PWS remains a healthy segment, but efficiency is now eroding and requires strategic attention.
- SPWS (Super Power Sellers) remains the marketplace's dominant GMV engine but is showing early signs of weakening. They created ~5.81M products/year (~484k/month) and generated ~$186M GMV/year (~$15.5M/month). GMV peaked in 2024 and dropped 25% in 2025, while product creation and merchant counts continued to rise. GMV per merchant fell from $30.2k → $28.7k → $26.3k, while GMV per product dropped sharply from $46 → $36 → $21, indicating efficiency erosion. Despite this, SPWS remains the highest-value segment by far and the core of marketplace revenue.
- Inactive merchants produce significant catalog volume but generate almost no revenue. On average, the segment created ~39.1M products per year (~3.26M/month), while average yearly GMV was only ~$20.7k/year (~$1.7k/month). GMV per product is essentially $0.00001–0.00002, and GMV per merchant is below $1 in two of three years. The trend shows decreasing product creation (2024 → 2025) and minimal improvement in GMV despite a massive number of merchants. This segment introduces large catalog noise with negligible commercial benefit.

YEAR_CREATED	SEGMENT	PRODUCT_COUNT	CREATION_MERCHANTS	PCT_PROD_OF_YE

2023-01-01 00:00:00.000		4019	2	0.00767919587
2024-01-01 00:00:00.000		7063	2	0.01174199796
2025-01-01 00:00:00.000		15398	3806	0.02730888916
2023-01-01 00:00:00.000	Inactive	40293503	2887923	76.989724265
2024-01-01 00:00:00.000	Inactive	42608952	2899896	70.835937656
2025-01-01 00:00:00.000	Inactive	34488097	2644119	61.165840918
2023-01-01 00:00:00.000	PWS	2215977	2870	4.234118295
2024-01-01 00:00:00.000	PWS	3044594	4263	5.06153427
2025-01-01 00:00:00.000	PWS	3190070	5672	5.657700224
2023-01-01 00:00:00.000	Regular	5785915	30912	11.055281058
2024-01-01 00:00:00.000	Regular	8560213	49191	14.231063801
2025-01-01 00:00:00.000	Regular	11219646	90440	19.898432853
2023-01-01 00:00:00.000	SPWS	4036795	1346	7.713197186
2024-01-01 00:00:00.000	SPWS	5930781	1880	9.859722275
2025-01-01 00:00:00.000	SPWS	7471360	2165	13.250717116


SEGMENT
null
Inactive
PWS
Regular
SPWS
Sum of PCT_PROD_OF_YEAR
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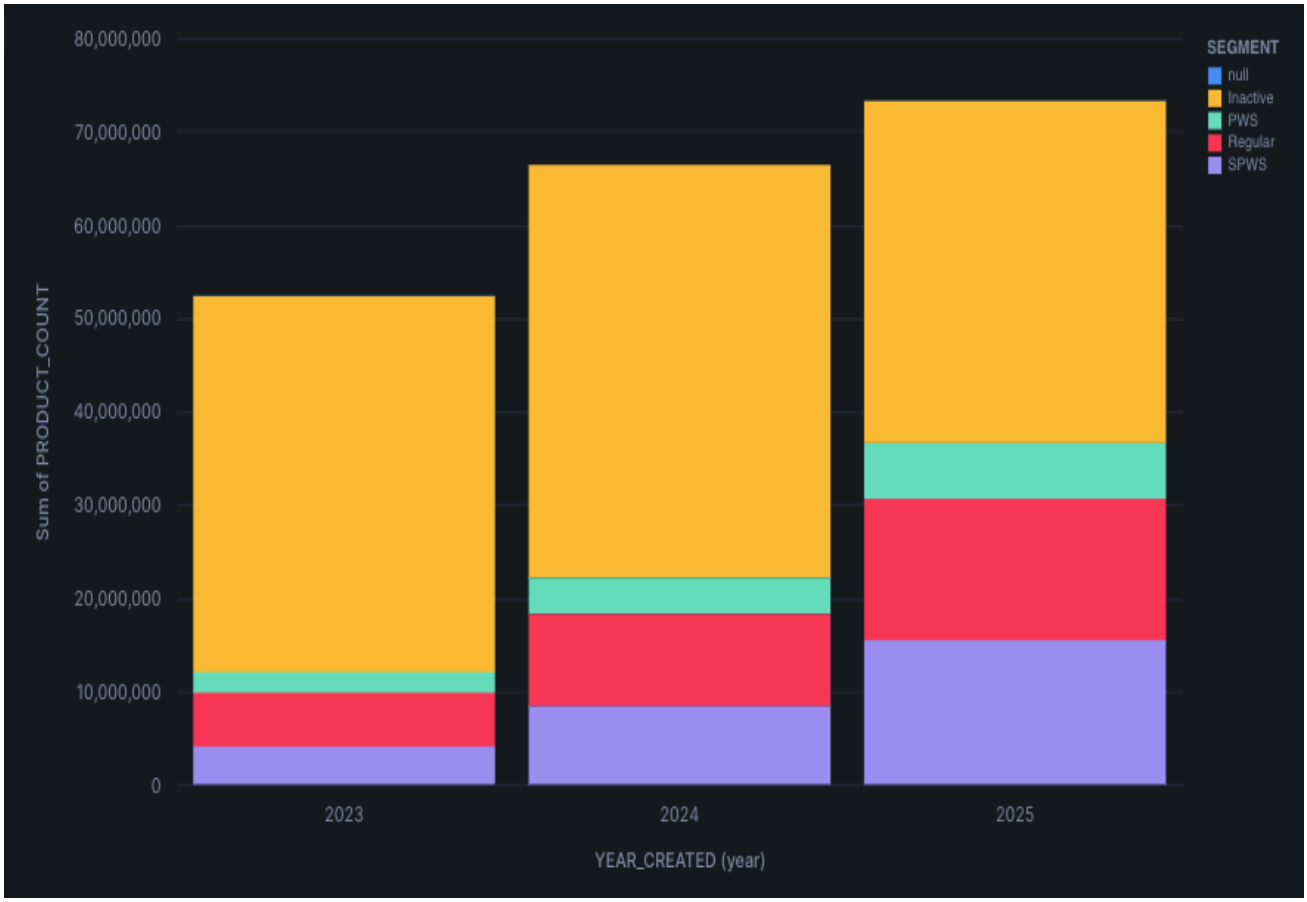
100
90
80
70
60
50
40
30
20
10
0
2023
2024
2025
YEAR_CREATED (year)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2024
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YEAR_CREATED (year)
SEGMENT
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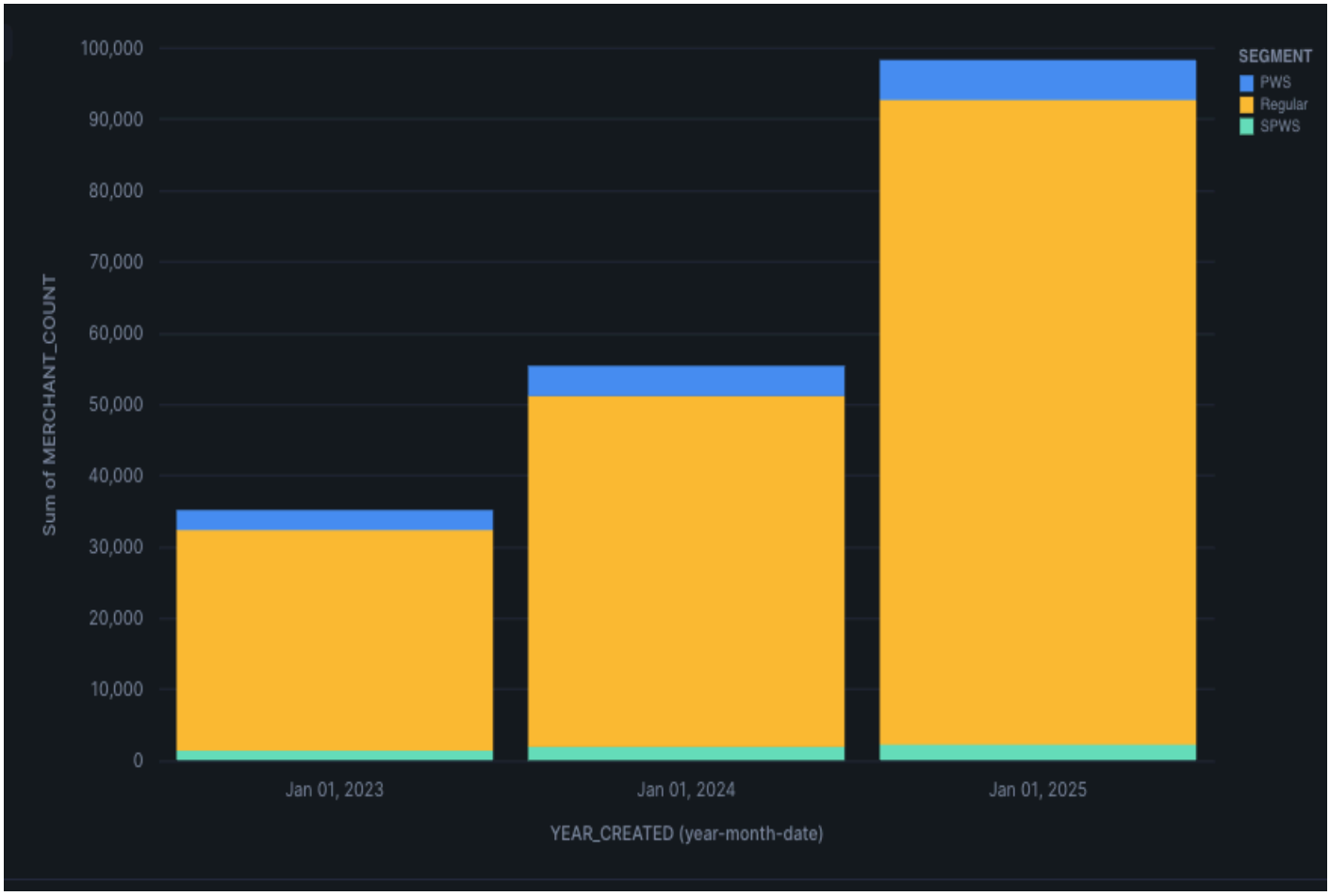
Regular
SPWS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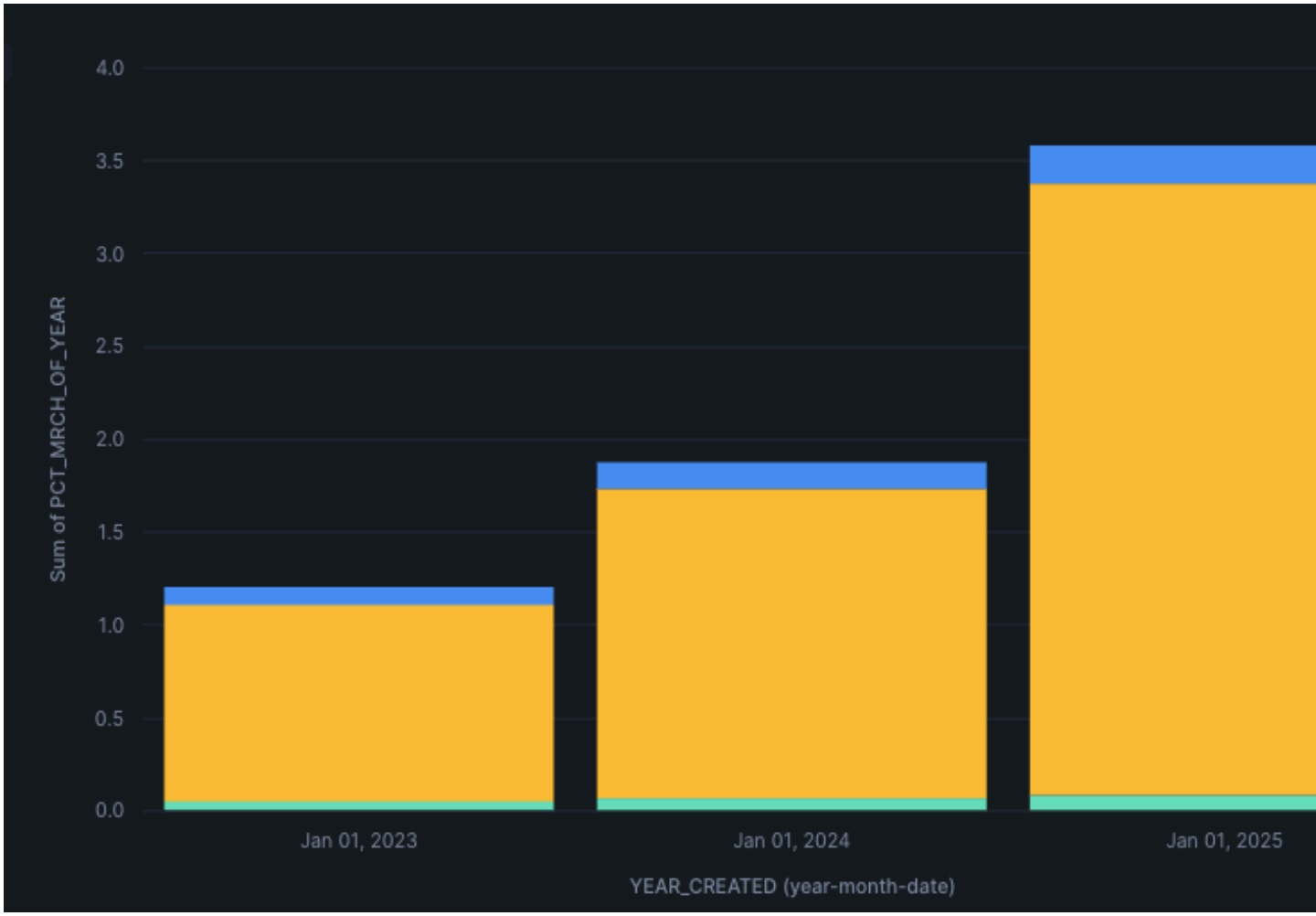
Jan 01, 2024
Jan 01, 2025
YEAR_CREATED (year-month-date)
SEGMENT
PWS
Regular
SPWS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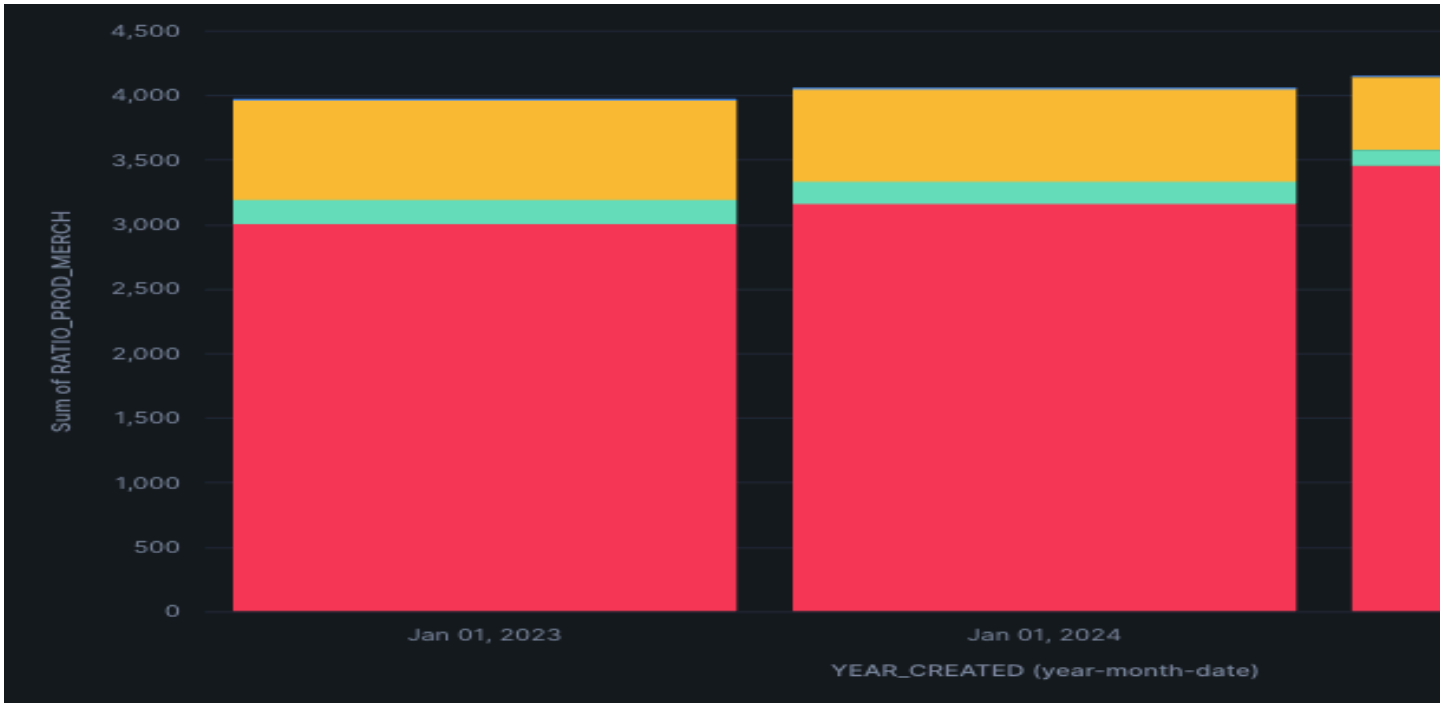
Sum of PCT_MRCH_OF_YEAR
Jan 01, 2023
Jan 01, 2024
Jan 01, 2025
YEAR_CREATED (year-month-date)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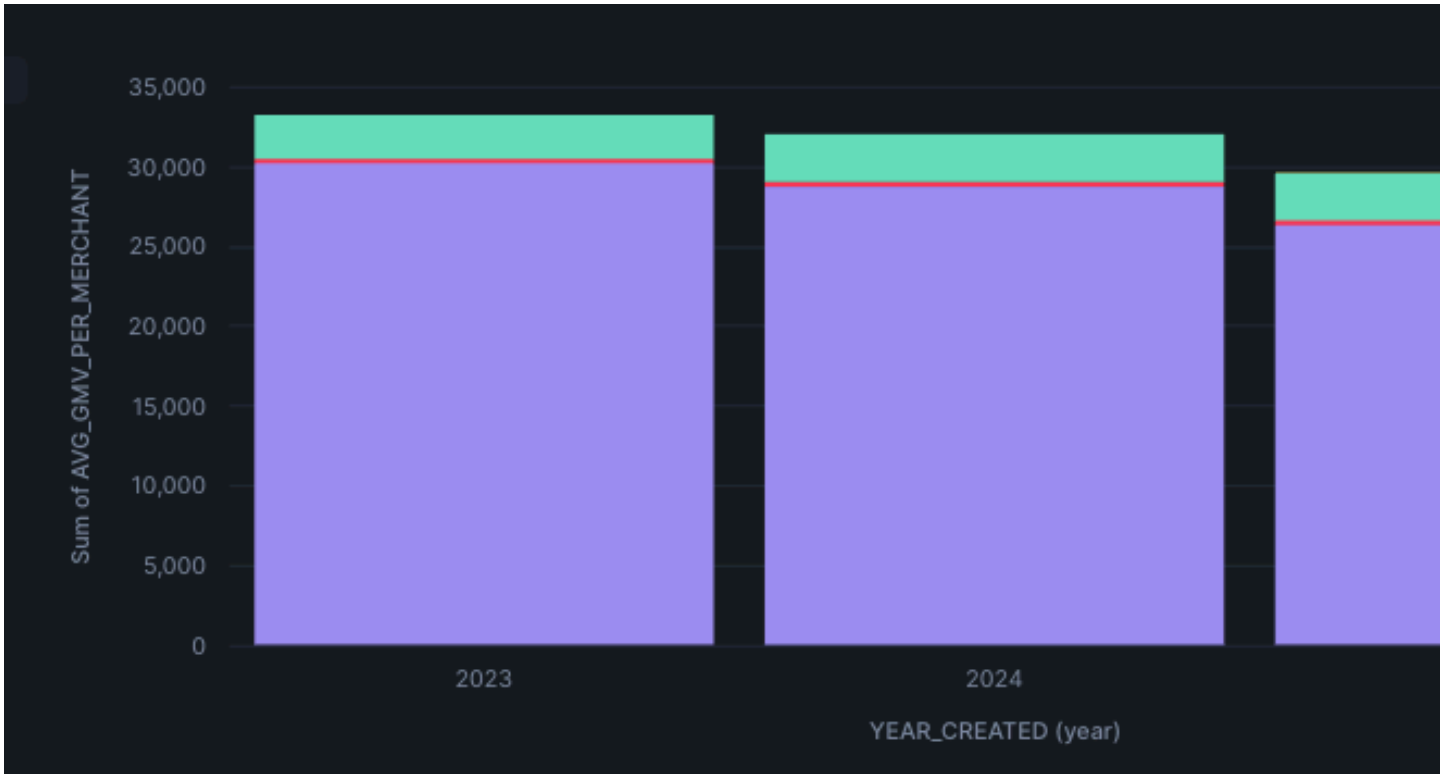
Jan 01, 2024
YEAR_CREATED (year-month-date)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30,000
25,000
20,000
15,000
10,000
5,000
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Sum of AVG_GMV_PER_MERCHANT
2023
2024
YEAR_CREATED (year)

100
90
80
70
60
50
40
30
20
10
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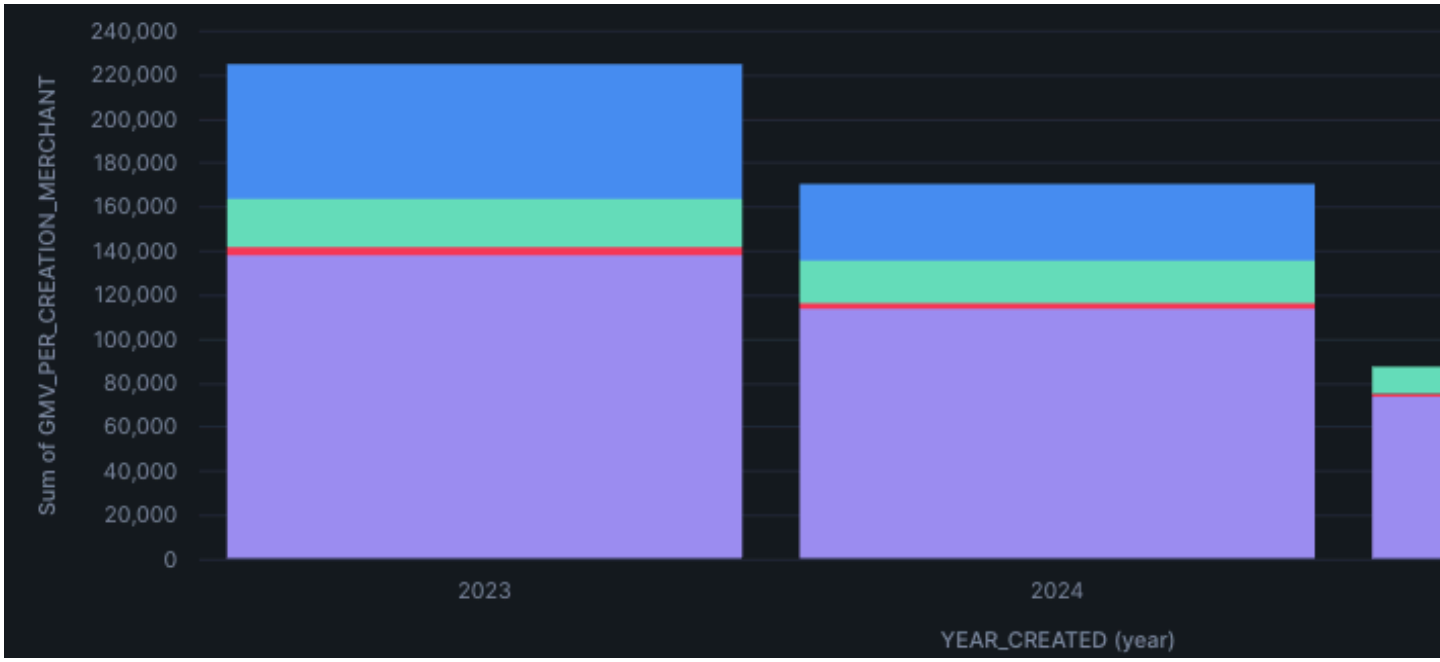
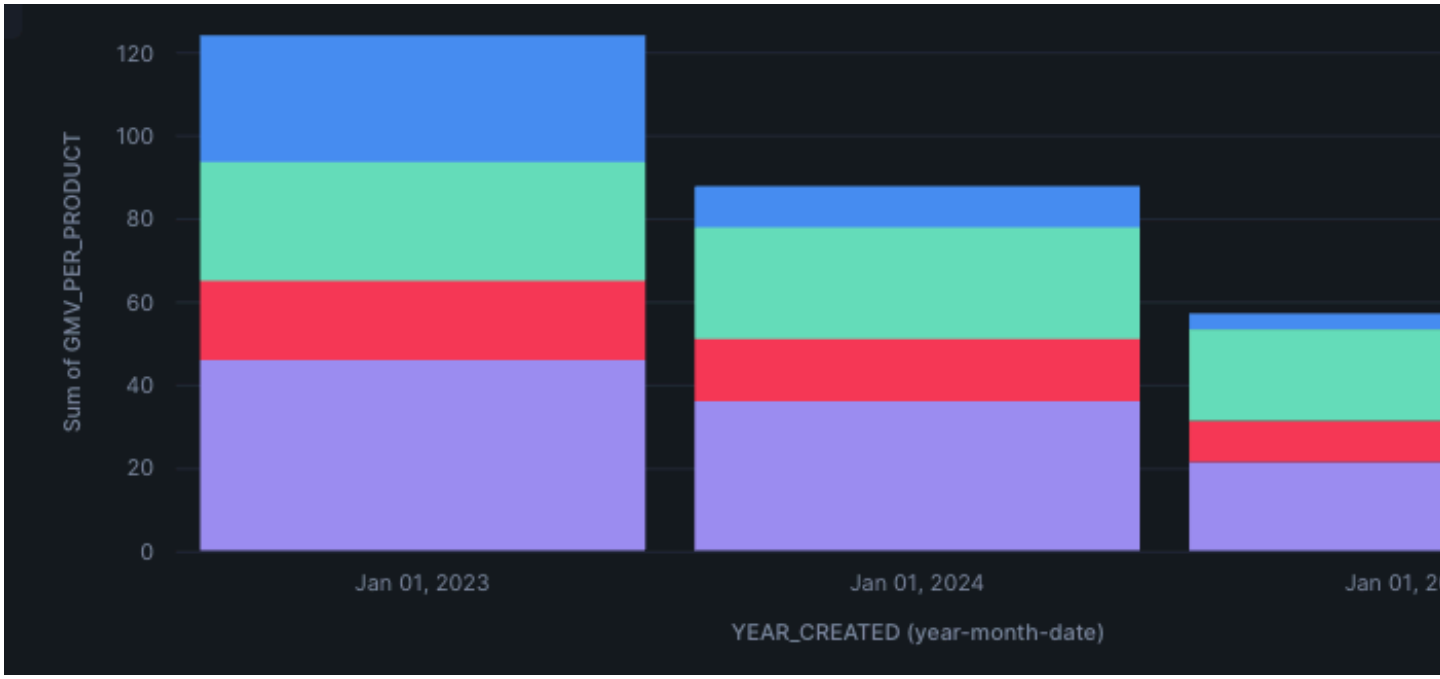
Sum of PCT_GMV_PER_SEGMENT
2023
2024
2025
YEAR_CREATED (year)





Trends per segment

PWS (Power Sellers)

This segment is growing steadily and is GMV-productive.

Trends

Metric	2023 → 2024	2024 → 2025
Product Count	+37%	+4.8%
Creation Merchants	+48%	+33%
GMV	+29%	–14%
GMV Merchants	+18%	–13%
GMV per Product	–6%	–18%
GMV per Merchant	+9%	–0.33%

Interpretation

- 2024 was a strong expansion year (more products + more merchants).
- 2025 shows a GMV contraction (–14%) despite more products.
- Efficiency declines: GMV/product and GMV/merchant falling.

PWS is growing supply faster than GMV → oversupply → dilution.

REGULAR

This is the largest segment by product count.

Metric	2023 → 2024	2024 → 2025
Product Count	+48%	+31%
Creation Merchants	+59%	+84%
GMV	+17%	–13%
GMV per Product	–21%	–34%
GMV per Merchant	+9.5%	–57% (!)

Interpretation

- Huge increase in new sellers in 2025, but GMV dropped.
- GMV productivity collapsed.
- This segment is becoming highly inefficient.

Regular segment is experiencing strong supply growth + GMV stagnation → sharp efficiency drop.

SPWS (SuperPower Sellers)

This is the highest-value segment with pure dominance in GMV share.

Metric	2023 → 2024	2024 → 2025

Product Count	+46.8%	+12.1%
Creation Merchants	+39.7%	+15%
GMV	+15%	–25%
GMV Merchants	+21%	–18%
GMV per Product	–22%	–41%
GMV per Merchant	–4%	–8%

Interpretation

- SPWS still owns half of all GMV (largest share).
- BUT 2025 shows sharp shrinkage in GMV and efficiency.
- GMV per product has fallen 61% in two years.

SPWS remains the GMV engine but is losing efficiency fast.

INACTIVE

Merchants no longer active in marketplace activity.

Trends

Metric	2023 → 2024	2024 → 2025
Product Count	+5.8%	–19.1%
Creation Merchants	+0.4%	–8.8%
GMV	+91%	–94% (!)
GMV Merchants	–14%	+143% (but still tiny)
GMV per Product	+81%	–92%
GMV per Merchant	+91%	–93%

Interpretation

- Product volume is enormous but GMV is virtually zero.
- 2025 shows a collapse in residual GMV (–94%).
- This segment is increasingly irrelevant economically.

Inactive merchants produce products but generate no GMV—likely clutter/noise.

UNASSIGNED SEGMENT (“blank segment”)

A tiny, anomalous bucket: only 2 merchants in 2023–2024 and then 3806 in 2025.

Trends

Metric	2023 → 2024	2024 → 2025	Notes
Product Count	+76%	+118%	Driven by large jump in merchants in 2025.
Merchants	Flat (2)	Explodes to 3806	Likely a data quality change.
GMV	–43%	–15%	GMV is trivial compared to other segments.
GMV per Product	+224%	–61%	Volatile due to tiny GMV.
GMV per Merchant	+10%	Stable	Still extremely low.

This segment is not meaningful until 2025, and even then GMV is insignificant.
Likely should be excluded or remapped.

Activation funnel (2nd Dec)

See also 2026 Strategy data

SETUP

- *Definition:* connected store + 3 published products within 7 days
- Returned metrics:
  - % merchants reaching SETUP
  - average days to SETUP

AHA

- *Definition:* first sale via integration within 30 days
- Metrics:
  - % merchants reaching AHA
  - average days to AHA

ACTIVATION

- *Definition:* first successful order sent to production
- Metrics:
  - % merchants activated

    - average days to activation

SCALEUP

- *Definition:* 10 orders via integration within 60 days
- Metrics:
    - % merchants reaching SCALEUP
    - average days to SCALEUP